Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Microprocessor and Computer Architecture**

Code No: Semester: **2**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. Explain the components of SAP 1 computer with its block diagram. How is it different from SAP 2 computer?

2. What are different types of instructions available in 8085 microprocessor? Write a program to find smallest element of an array: (Assume all the necessary data and memory location).

3. Explain the instruction format of basic computer. Differentiate between RISC and CISC computer.

4. Explain arithmetic micro-operations with suitable examples.

5. What is micro-program? Write symbolic microprogram for FETCH operation.

6. Explain hardware implementation and algorithm for subtraction and addition of signed magnitude data with suitable diagram.

7. Explain about memory hierarchy in computer.

8. Explain the basic architecture of microprocessor based system.

9. What is flag? What are different flags available in 8085 microprocessor?

10. Explain arithmetic pipeline with suitable example.

11. What is the function of I/O interface? Differentiate between Isolated I/O and memory mapped I/O.

12. Write short notes on (any two):a. Control ROMb. Magnetic Diskc. Flynn’s Classification